LAB 05 UTP CABLING

SECR1213 NETWORK COMMUNICATIONS

1.0 Introduction

Common Ethernet UTP network cable are straight and crossover cable. This Ethernet UTP network cable is made of 4 pair high performance cable that consists twisted pair conductors that used for data transmission. Both end of cable is called RJ45 connector. Straight and crossover cable difference is each type will have different wire arrangement in the cable for serving different purposes.

2.0 Objectives

- Exploring the EIA/TIA-568A/B standards for connecting the RJ45 connector and the UTP network cable.
- Get the initial skills of crimping and testing UTP network cable

3.0 Initial Task

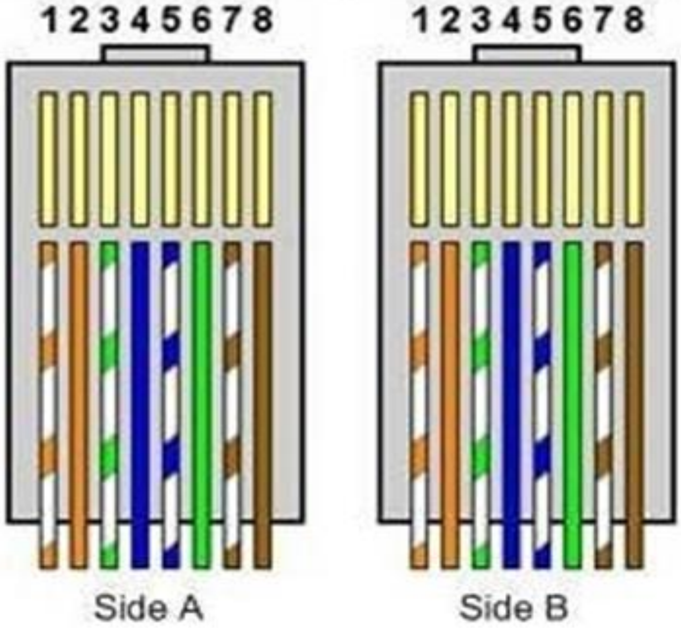
- Study of technical data and designs of EIA/TIA-568A/B, UTP network cable, RJ45.
- View short videos about handmade UTP network cable.

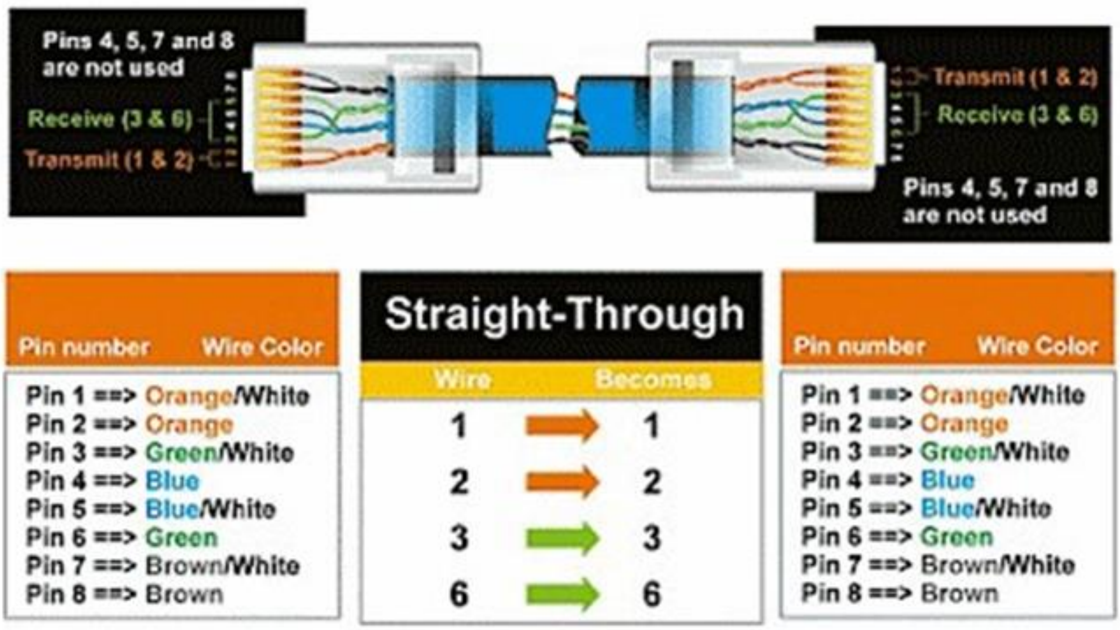
4.0 Straight UTP Cable

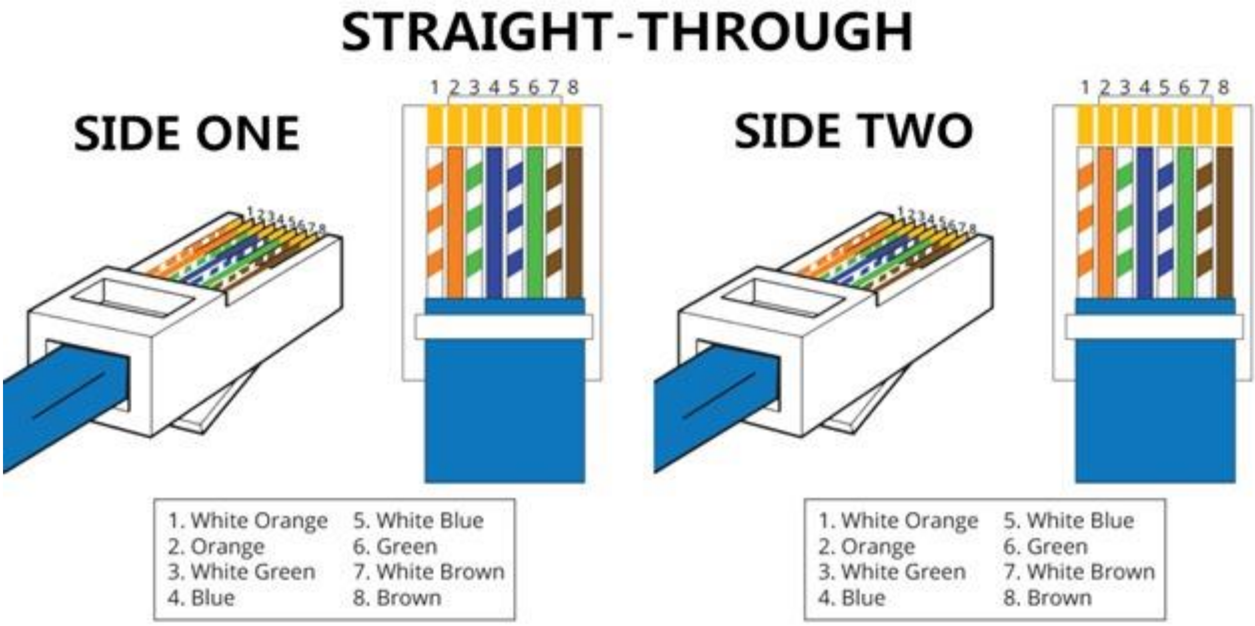
You usually use straight cable to connect different type of devices. This type of cable will be used most of the time and can be used to:

- Connect a computer to a switch/hub's normal port.
- Connect a computer to a cable/DSL modem's LAN port.
- Connect a router's WAN port to a cable/DSL modem's LAN port.
- Connect a router's LAN port to a switch/hub's uplink port. (normally used for expanding network)
- Connect 2 switches/hubs with one of the switch/hub using an uplink port and the other one using normal port.

Pin ID	Side A	Side B
1	Orange-white	Orange-white
2	Orange	Orange
3	Green-white	Green-white
4	Blue	Blue
5	Blue-white	Blue-white
6	Green	Green
7	Brown-white	Brown-white
8	Brown	Brown





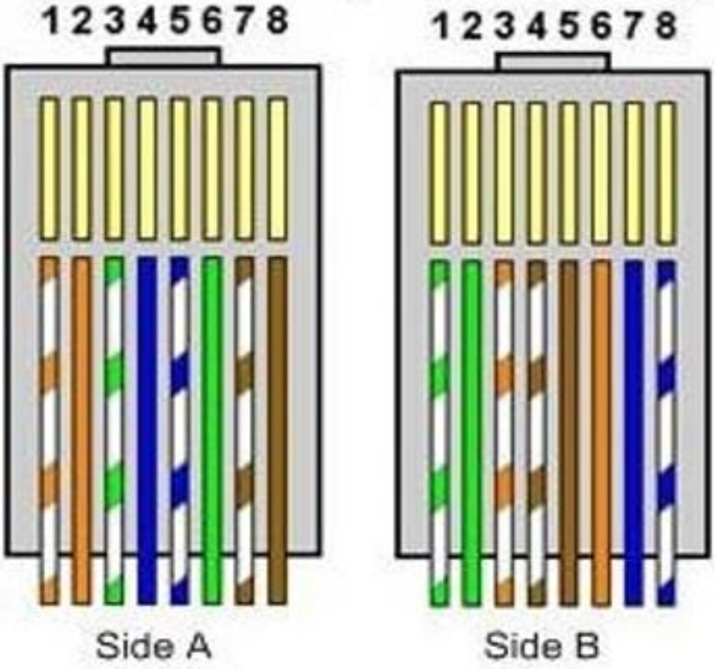


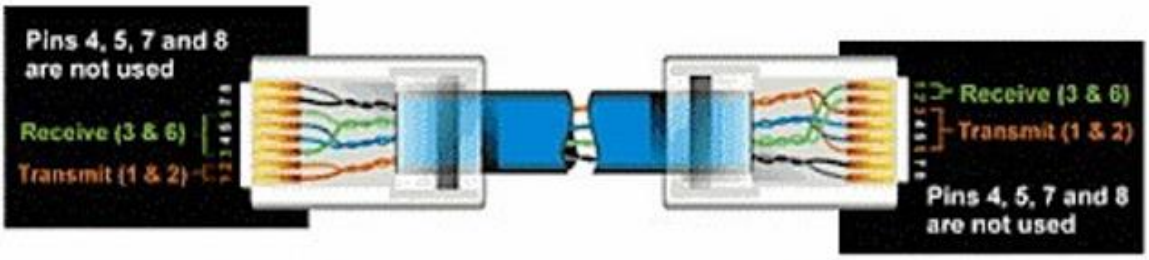
5.0 Crossover UTP Cable

Sometimes you will use crossover cable, it's usually used to connect same type of devices. A crossover cable can be used to:

- Connect 2 computers directly.
- Connect a router's LAN port to a switch/hub's normal port. (normally used for expanding network)
- Connect 2 switches/hubs by using normal port in both switches/hubs.

Pin ID	side A	side B
1	Orange-white	green-white
2	Orange	green
3	green-white	orange-white
4	blue	brown-white
5	blue-white	Brown
6	green	orange
7	brown-white	Blue
8	brown	blue-white





Pin number	Wire Color
Pin 1 ==>	Orange/White
Pin 2 ==>	Orange
Pin 3 ==>	Green/White
Pin 4 ==>	Blue
Pin 5 ==>	Blue/White
Pin 6 ==>	Green
Pin 7 ==>	Brown/White
Pin 8 ==>	Brown

Crossed-Over

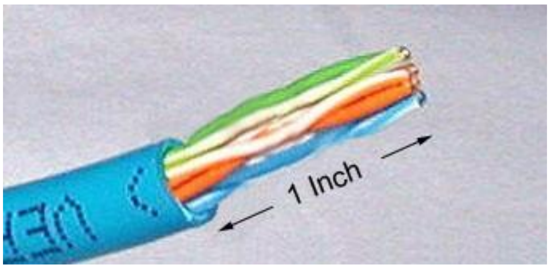
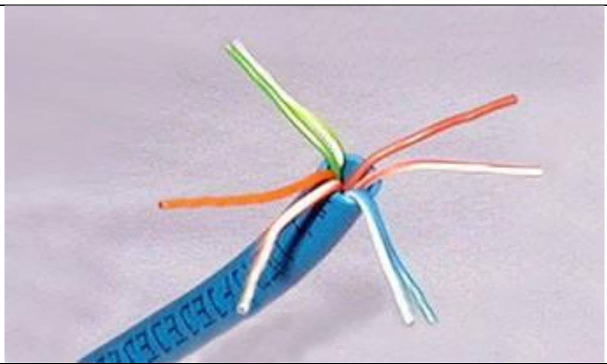
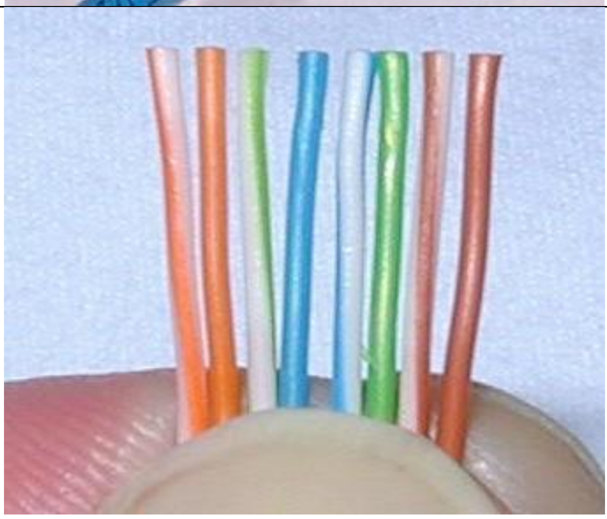
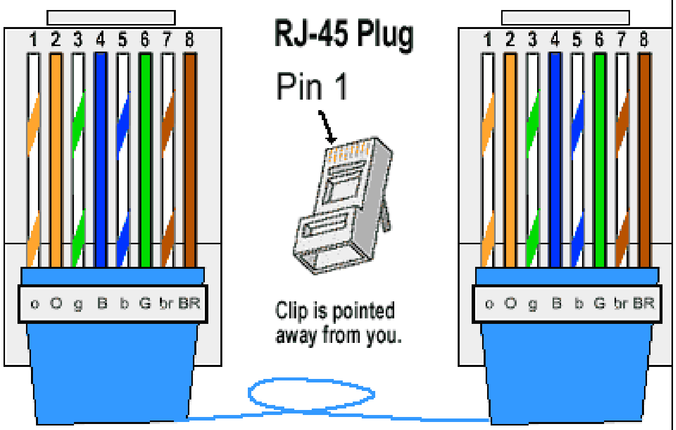
Wire	Becomes
1	3
2	6
3	1
6	2

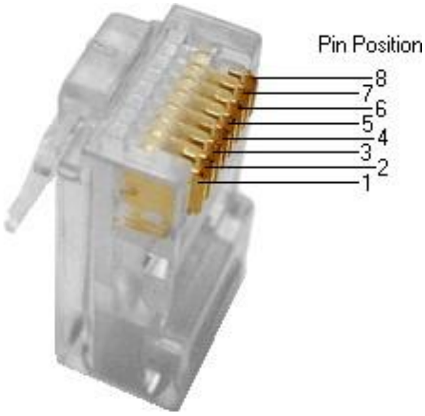
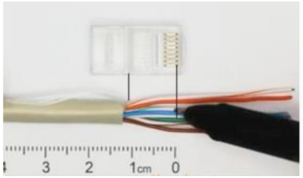
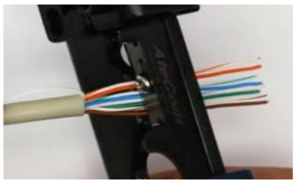
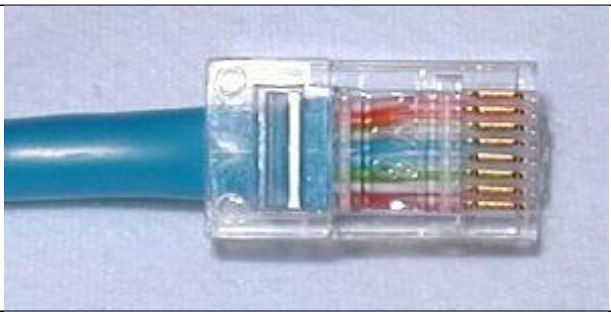
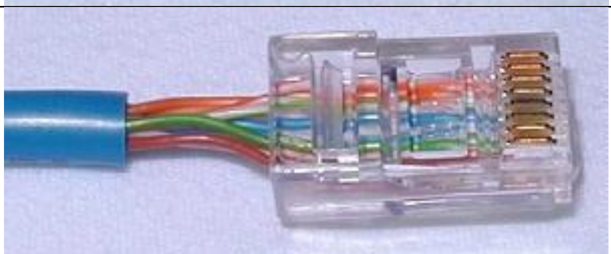
Pin number	Wire Color
Pin 1 ==>	Green/White
Pin 2 ==>	Green
Pin 3 ==>	Orange/White
Pin 4 ==>	Blue
Pin 5 ==>	Blue/White
Pin 6 ==>	Orange
Pin 7 ==>	Brown/White
Pin 8 ==>	Brown

6.0 Items Required

UTP Network Bulk Cable 	UTP RJ45 Crimpable Connectors 	UTP Plug Shields for RJ45 Connectors (optional) 
UTP Crimping Tool for RJ-45 (Crimper) 	UTP Cable Stripper 	UTP Cable Tester for RJ-45 

<u>7.0 Straight UTP Cable and RJ45 Installation Steps</u>

Step		Description
1		Use cable stripper clean the outer shell about **1 inch** (2.5 cm) from the one end of the cable. The crimping tool has a razor blade that will do the trick with practice. To do this: • insert the end of the cable into the rear round hole of the crimper • clamp the ends of the crimper and rotate the outer shell 
2		Unwind and pair the similar colors.
3		Pinch the wires between your fingers and straighten them out as shown. The color order is important to get correct. 

		
4	 OR 	Use crimping tool or scissors to make a straight cut across the 8 wires to shorten them to **1/2 inch** (1.3 cm) from the cut sleeve to the end of the wires.  
5a		Carefully push all 8 unstripped colored wires into the connector. TRUE WAY - Note the position of the blue plastic sleeve. Also note how the wires go all the way to the end.
5b		TRUE WAY - A view from the top. All the wires are all the way in. There are no short wires.
5c		WRONG WAY - Note how the blue plastic sleeve is not inside the connector where it can be locked into place. The wires are too long. The wires should extend only 1/2 inch from the blue cut sleeve.

5d		WRONG WAY - Note how the wires do not go all the way to the end of the connector.
6		Crimping the cable, carefully place the connector into the UTP Crimper and cinch down on the handles tightly. The copper splicing tabs on the connector will pierce into each of the eight wires. There is also a locking tab that holds the blue plastic sleeve in place for a tight compression fit. When you remove the cable from the UTP crimper, that end is ready to use.
7		Repeat all the steps on the other end
8		If you have a UTP tester, make sure to test the cables before installing them.

8.0 Straight UTP Cable and RJ45 Video Guide

URL Link: https://www.youtube.com/watch?v=NWhoJp8UQpo